

OLIMEXINO-STM32F3

User Manual

Document revision 3.0 September 2025

www.olimex.com

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What is OLIMEXINO-STM32F3

OLIMEXINO-STM32F3 is an industrial grade -40+85C operating temperature, Open Source Hardware board with STM32F303RCT6 main chip. The chip has 256 KB of flash memory and 48 KB of RAM. The board works with powered off a Li-Po battery without an external supply, and switching between USB to external to Li-Po battery power supply is seamless and automatic. If USB is connected the board gets powered by the USB, if external power is applied to the power jack – it has priority over the USB and battery and the board will be powered by the external power supply, if both external power supply and USB power supply are missing but a battery is connected the board would automatically get powered by the LiPo battery.

In STM32F3 series of microcontrollers the CAN and the USB can work at the same time.

Notice that this guide does not apply for OLIMEXINO-GD32F3!

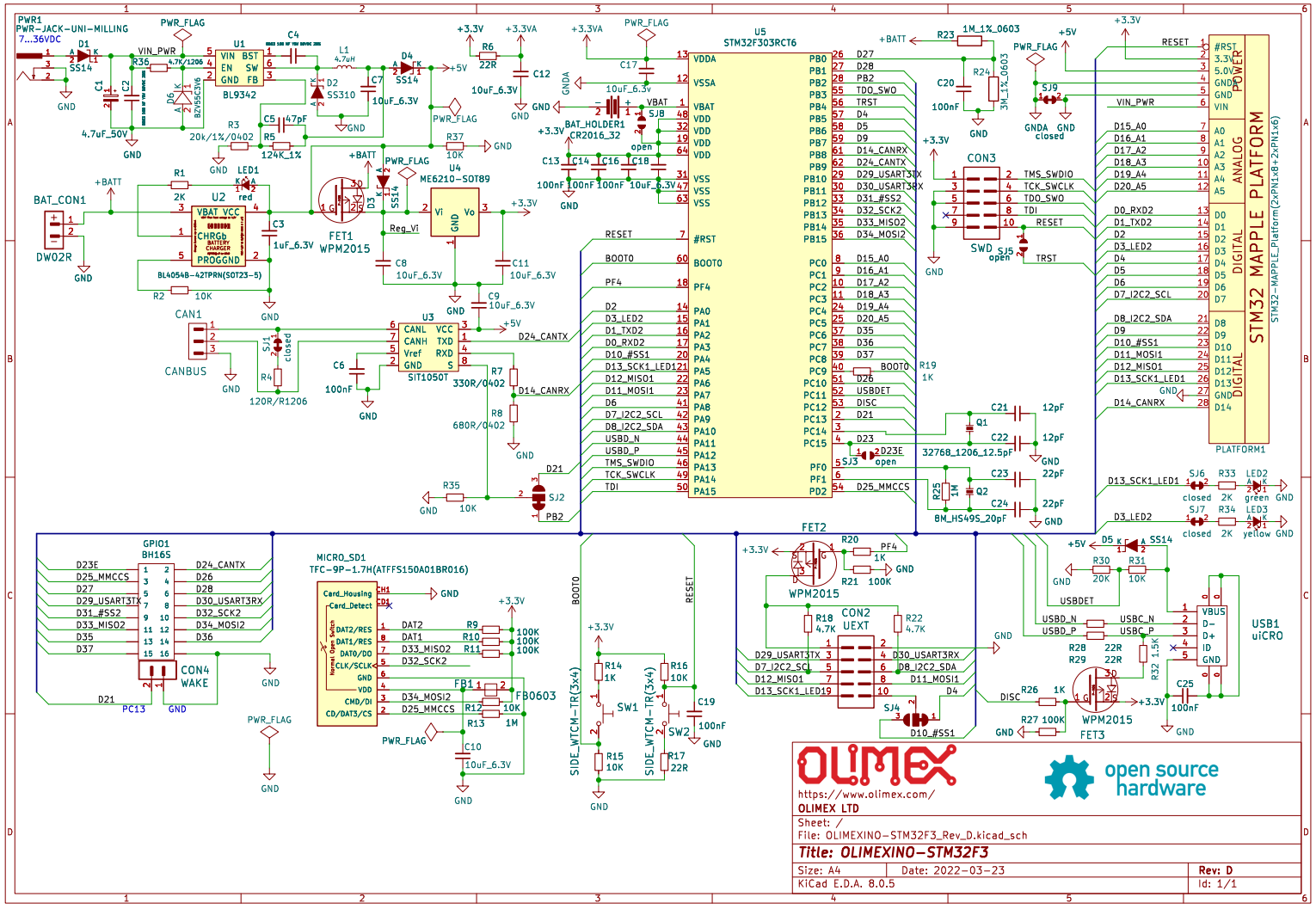
OLIMEXINO-STM32F3 has following features:

- STM32F303RCT6 256KB flash, 40KB RAM
- Industrial grade (-40+85)°C operating temperature
- Micro USB connector for powering and programming
- Power jack (5.5mm x 2mm) for external power supply (4.2-40)V DC
- Li-Po battery charger and connector
- Micro SD card
- RTC 32.768kHz with CR2032 battery connector
- Bootloader button
- Reset Button
- CAN driver and connector
- UEXT connector
- Arduino connectors
- Extra 16 pin GPIO connector
- 10-pin JTAG connector for programming and debugging

Product page:

https://www.olimex.com/Products/Duino/STM32/OLIMEXINO-STM32F3/open-source-hardware

Hardware schematic, revision D



Programming considerations

OLIMEXINO-STM32F3 can be programmed in a number of ways. It can be programmed via JTAG or SWD tool using its 10-pin JTAG connector; it can be programmed via the built-in USB or serial bootloader. A lot of software tools can be used. For simple programming and re-programming tasks maybe consider “STM32CubeProgrammer” stand-alone programmer software. For development work again a lot of software tools can be used depending on the goals and proficiency of the user. We would provide information about how you can program the board with Arduino IDE:

Arduino IDE installation

The OLIMEXINO-STM32F3 board is natively supported in newest releases of the STM32's Arduino core package.

Notice! Since core release 2.8.0, only Arduino IDE 2 is supported.

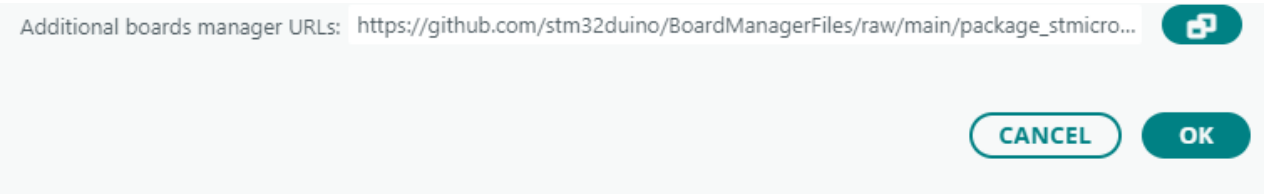
This means that it is recommended to install Arduino IDE 2.x.x first. Latest Arduino IDE can be downloaded from here:

https://www.arduino.cc/en/software/

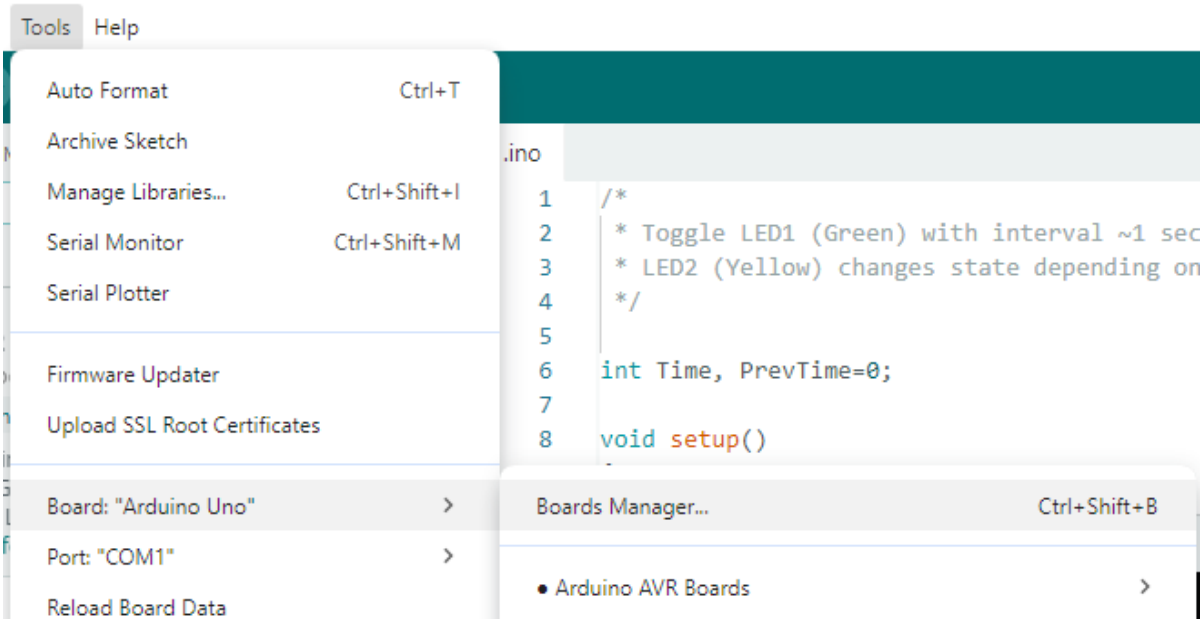
1. Once you have Arduino IDE 2.x.x installed, start it and navigate to Files→ Preferences and in “Additional Boards Manager URLs” field append the following link:

https://github.com/stm32duino/BoardManagerFiles/raw/main/package_stmicroelectronics_index.json

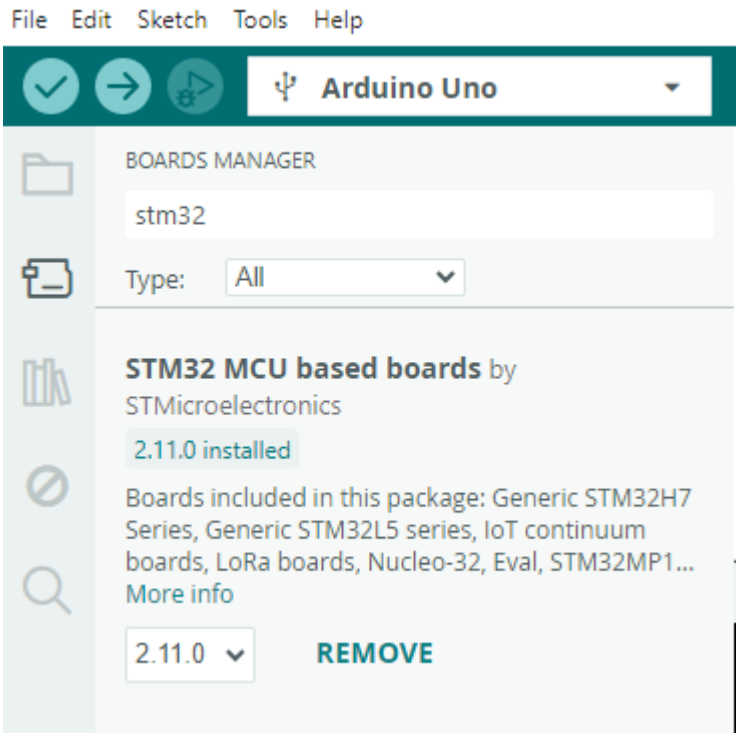
as shown below:



2. Then navigate to Tools→ Boards → Board Manager



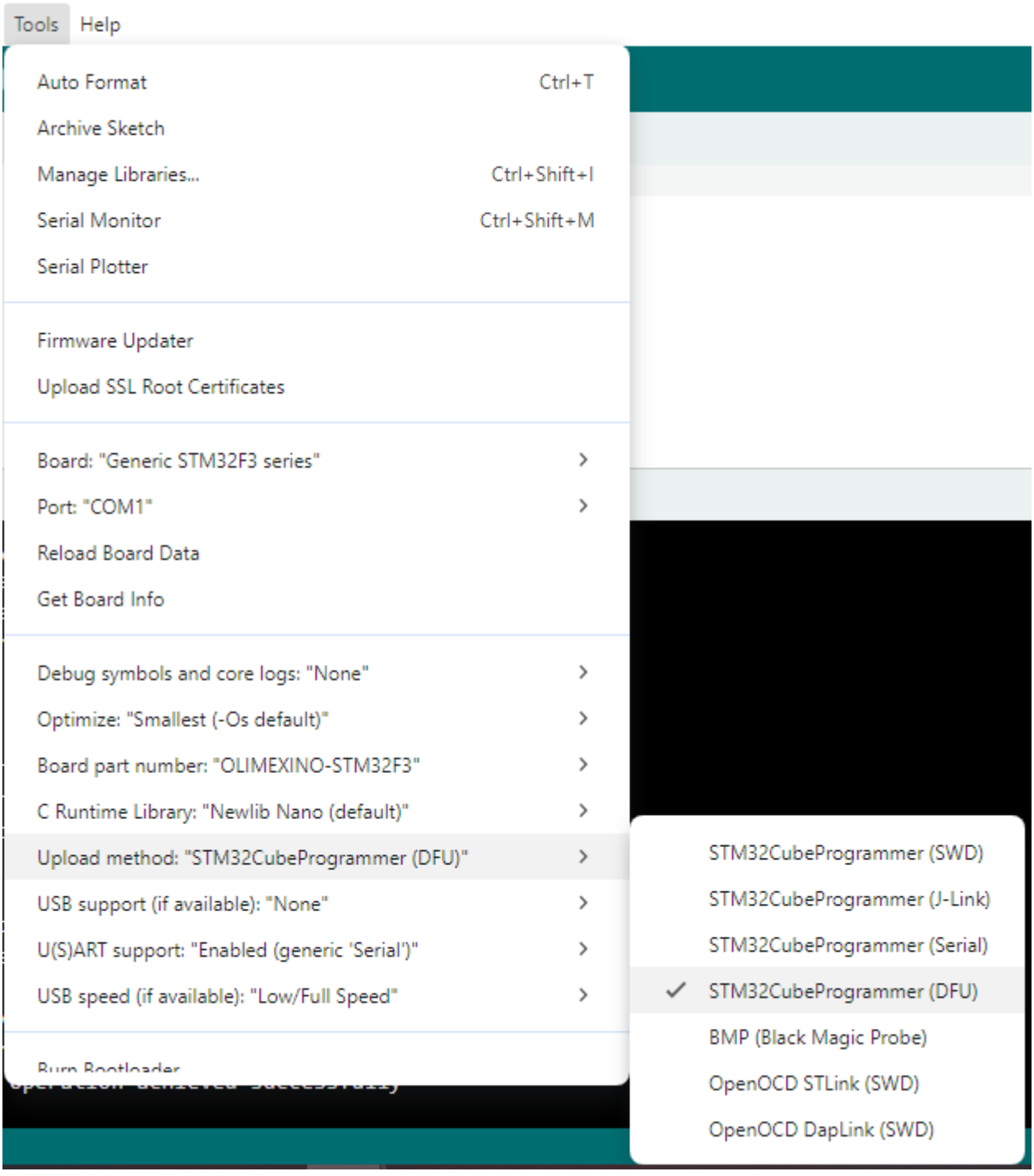
3. Type “stm32” to filter out the available packages then install the latest one made by STMicroelectronics. Wait until installations completes. It should look similar to this:



4. I want to upload the demo via USB DFU. And if you wish to upload via SWD, Serial, or DFU you need to install [STM32 Cube Programmer] – make sure to download and install it.

5. Restart Arduino IDE.

6. Navigate to Tools -> Board -> STM32 MCU based boards -> "Generic STM32F3 series" and then from Tools -> Board Select Part -> OLIMEXINO-STM32F3. Under upload method remember to select DFU, which is the microUSB method (alternatively you can also serial method but that requires Serial-USB adapter attached to the GND, RX, TX pins at the UEXT connector). Follow the image below:



7. Put the board in USB DFU mode – this is done by button manipulation – press and hold down button SW1 then press and release button SW2, then release button SW1. The board should now be recognized as “STM32 bootloader”.

8. Load one of the examples from our GitHub page: https://github.com/OLIMEX/OLIMEXINO-STM32F3/tree/master/SOFTWARE

I test with Olimexino-STM32F3_GPIO. Compile and upload. Green LED should blink, pressing SW1 should turn on yellow LED.

If you encounter issues with the installation make sure to also check the official installation instructions:

https://github.com/stm32duino/Arduino_Core_STM32/wiki/Getting-Started

How to use STM32CubeProgrammer

It is a popular programmer software if you already have a binary and just want to program the board (or read or erase it).

1. Download, install, and run STM32CubeProgrammer

2. Connect the board to your computer using the USB1 micro USB connector

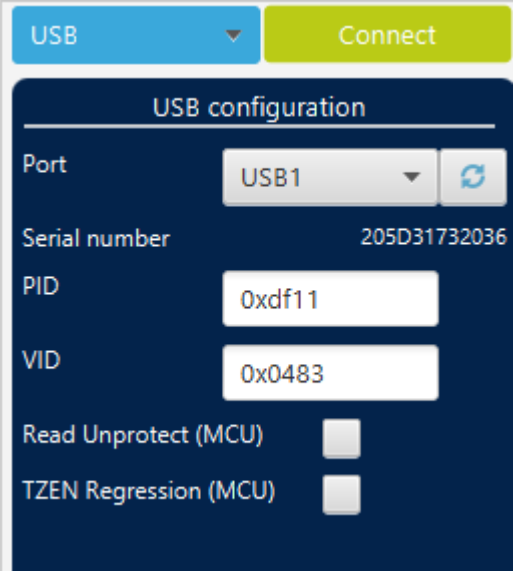
3. Put the board in USB DFU mode – this is done the following way:

- press and hold button SW1

- press and release button SW2

- release button SW1

4. Select USB connection from the download menu and the board should be found by STM32CubeProgrammer. Click “Connect”. Refer to the picture below.



5. Now you can read, or program, or erase the connected board.

Software examples and libraries

You can find our work here:

https://github.com/OLIMEX/OLIMEXINO-STM32F3/tree/master/SOFTWARE

In GitHub software folder there are examples on how to use the GPIOs (LEDs and button), I2C, SPI, CAN, UART, SD-card, and more.

Document revision history

Revision 3.0 September 2025

- Updated Arduino IDE instructions for latest software releases
- Updated schematic to latest hardware revision

Revision 2.0 September 2022

- Updated installation instructions
- Updated schematic to latest one
- Minor spelling and design improvements

Revision 1.0 July 2020

- Initial release of the document